

‡ CLINICIAN REFERENCE

KDIGO 2020 ALIGNED

Isolated Ultrafiltration, Sodium & UF Ramping

A clinician's selection & prescription guide

A mechanism-driven selection and prescription framework for hemodynamically vulnerable HD patients.

Published: Jun 27, 2026 **References:** 20 **Audience:** Clinicians (IM · Nephrology · dialysis nurses with clinical authority)

Read time: ~26 min

Clinical use only. These techniques are *hemodynamic stabilization tools*, not adequacy tools. Their value is concentrated in an identifiable minority of HD patients; in everyone else they add cost, complexity, prescribing-error risk, and the documented harms of sodium loading and masked volume overload. Use this guide to identify that subset — and to *avoid* these prescriptions in everyone outside it.

BEST USED WHEN — AT A GLANCE

S1 · PHYSIOLOGY

The Shared Physiology — Why Blood Pressure Fails During Fluid Removal

Iso-UF, sodium ramping, and UF ramping all act on the same four-axis cascade. Selecting the right tool starts with naming which axis is failing in this patient.

Every intradialytic-hypotension (IDH) event is the visible end of an unfavourable arithmetic: instantaneous fluid removal exceeds the patient's instantaneous ability to compensate. The three techniques in this guide are different mechanical answers to the same physiologic question — *how do we keep effective circulating volume and vascular tone intact while net body water falls?*^{2 (#ref-2)}

Axis 1 — Plasma refilling vs UFR mismatch

Fluid leaves the intravascular compartment at the ultrafiltration rate (UFR). It is replaced from the interstitium at the plasma-refilling rate, which is finite and falls as the session progresses (interstitial hydrostatic pressure drops, oncotic pressure rises, lymphatic return saturates). When UFR exceeds refilling, relative blood volume falls, cardiac filling falls, and IDH follows.^{2 (#ref-2)} This is the dominant mechanism behind weight-based UFR thresholds ($\approx 10\text{--}13$ mL/kg/hr).^{14 (#ref-14), 15 (#ref-15)}

Axis 2 — Osmolar/tonicity collapse

Diffusive removal of urea and other small osmolytes during conventional HD lowers plasma osmolality during the session. Water shifts intracellularly (impairing refilling) and the osmotic drive to vasoconstriction is blunted. This is the *specific* axis iso-UF and sodium profiling were designed to support: iso-UF avoids diffusive solute loss entirely during the UF block;^{5 (#ref-5)} Na ramping replaces lost tonicity early when the UFR is highest.^{6 (#ref-6), 9 (#ref-9)}

Axis 3 — Autonomic and vascular tone failure

Adequate refilling and adequate *tonicity* still require the vasculature to recruit compensatory venoconstriction and shift blood out of the splanchnic bed. Long-standing diabetic autonomic neuropathy, elderly age, polypharmacy with antihypertensives, and severe diastolic dysfunction all blunt this response.^{3 (#ref-3)} A patient who cannot vasoconstrict will not be rescued by any single hemodynamic technique — but iso-UF's preserved peripheral vascular resistance (PVR) is one of the few in-session interventions that addresses this axis.^{4 (#ref-4)}

Axis 4 — Cardiac reserve

Diastolic dysfunction, LVH, severe aortic stenosis, and low-output states are all preload-dependent. As preload falls during UF, these hearts cannot augment output. The Pirkle and KDIGO datasets agree: cardiac stress is a function of *average* UFR, not its distribution. Profiling alone does not protect a stunned myocardium if mean UFR is too high.^{13 (#ref-13), 14 (#ref-14)}

MECHANISM SUMMARY — WHAT EACH TOOL DOES

Iso-UF protects axes 2 and 3 (preserved osmolality → preserved PVR). **Na ramping** protects axis 2 (delivered tonicity). **UF profiling** protects axis 1 (UFR matched to falling refill curve). None protects axis 4 — that requires lowering the *mean* UFR (more time, more frequency, or lower IDWG).

Why the subset matters — the end-organ ledger

Repeated IDH is not a per-session inconvenience. Each event delivers a microcirculatory hit to the myocardium (Burton's "myocardial stunning"), cerebral white matter, gut mucosa (with translocation/endotoxemia signals), the vascular access (with thrombosis risk), and any residual nephrons.^{18 (#ref-18)} A patient with ten IDH events per month is accumulating end-organ injury at a rate the dialysis prescription itself can modify. This is the population in which these techniques earn their place.



Figure 1 — the refilling, osmolality, and tone cascade. A four-axis mechanism schematic mapping IDH physiology to its hemodynamic-stabilization tools: UF ramping for refilling/UFR mismatch, iso-UF and sodium-balance-neutral ramping for tonicity collapse, iso-UF and cool dialysate for vascular tone failure, and time/frequency increase plus dry-weight reassessment for cardiac reserve limits.

Figure 1. The refilling / osmolality / tone cascade — four physiologic axes that govern IDH, with each axis mapped to its mechanism-matched stabilization tool. The magnified inset contrasts conventional HD (diffusive osmolyte loss → tonicity collapse → blunted vasoconstriction) with iso-UF (osmolytes retained → tonicity preserved → PVR maintained).

§2 · TECHNIQUE — ISO-UF

Isolated Ultrafiltration (Iso-UF / Sequential UF-then-HD)

Convective fluid removal without diffusion. Preserves plasma osmolality and peripheral vascular resistance — at the cost of zero clearance during the UF block.

Definition

Iso-UF is fluid removal by convection with dialysate flow off or with no diffusive gradient. Solute clearance during the UF phase is essentially nil. It is most commonly delivered as *sequential dialysis*: a discrete iso-UF block (front-loaded, when refill reserve is highest) is followed by a conventional HD block that delivers the diffusive dose.

Mechanism of benefit

Because no urea or small osmolytes are removed during the iso-UF block, plasma osmolality is preserved. Two consequences follow:

- The water content of the intravascular compartment is not diluted by an inward osmotic shift — **plasma refilling is more efficient.** ^{2 (#ref-2)}
- The osmotic drive sustaining vascular tone is preserved — **peripheral vascular resistance does not fall as steeply** as during simultaneous HD+UF. The 2025 hemodynamics study supports this as the observed correlate of iso-UF's benefit, although the upstream mechanism is incompletely resolved. ^{4 (#ref-4),5 (#ref-5)}

WHY ISO-UF IS PHYSIOLOGICALLY DIFFERENT

It is not "the same UF, with less stress." It is a **different hemodynamic event**: the tonicity-preservation pathway is closed in conventional HD+UF and open in iso-UF. This is the clinical justification for choosing it over simply slowing UFR.

What iso-UF does *not* do

There is no diffusive clearance during the UF block. Total session Kt/V is the HD-block Kt/V only. If treatment time is not extended, the diffusive dose must be compensated by higher blood-flow, dialysate-flow, or membrane KoA — and the realistic ceiling on these is finite. **State the adequacy trade-off explicitly in the order** (§7 worksheet).

Operational prescription notes

- **Sequence:** front-load the iso-UF block (refill reserve and tonicity drive both highest at session start), then HD block.
- **Duration / rate of the iso-UF phase:** no guideline-endorsed standard. The KDIGO Controversies Conference declined to endorse one. ^{1 (#ref-1)} Individualize; document rationale in the order.
- **Total session time:** sequential dialysis either extends the total session or compresses the clearance phase. Decide which, and consent the patient.
- **Anticoagulation:** two distinct fluid-only and dialysate-on phases — confirm heparin dose covers both intervals.
- **Vascular access flow:** blood-flow may be intentionally reduced in the iso-UF block to ease hemodynamics; reset for the HD block to recover Kt/V.

Figure 3. Sequential dialysis on a time axis. Front-loaded iso-UF block (no diffusion, preserved osmolality, PVR support) → HD block (full Kt/V dose, tapered UFR). The adequacy trade-off is real and lives in the order sheet.

EVIDENCE AT A GLANCE — ISO-UF

STRENGTH OF EVIDENCE

Small mechanistic/hemodynamic studies; one KDIGO consensus framing.

WHAT IS NOT PROVEN

Mortality or hard-outcome benefit.

WHAT IS PROVEN

Hemodynamic stability and PVR preservation in selected patients.¹
(#ref-1),⁴ (#ref-4)

PRINCIPAL HARM

Adequacy (Kt/V) loss if the HD block does not compensate; lengthens session.

S3 · TECHNIQUE — SODIUM RAMPING

Dialysate Sodium Profiling & Sodium Ramping

A tonicity-replacement strategy. Powerful when designed for net-zero diffusive Na — harmful when not.

Definition & taxonomy

Sodium ramping (a subset of sodium profiling) means setting the dialysate Na high at session start — typically **150–160 mEq/L** — and stepping or linearly decreasing it toward ≈ 140 mEq/L by session end.^{12 (#ref-12)} Profile shapes include:

- **Step** — discrete decrements (e.g. 155 → 145 → 140 over thirds of the session).
- **Linear descending** — continuous taper (the most common implementation).
- **Exponential** — front-loaded tonicity delivery, faster taper.
- **Ascending-descending (A/D)** — increasingly used in combined Na + UF profiling studies.^{11 (#ref-11)}

Mechanism

Early high dialysate Na delivers a positive diffusive Na flux into plasma during the steepest UF interval. The plasma tonicity that would otherwise fall (from urea/osmolyte removal) is supported, the inward water shift is blunted, refilling is preserved, and the osmotic component of vasoconstrictor tone is maintained.^{9 (#ref-9),10 (#ref-10)} The descending limb aims to return the net diffusive Na balance toward zero by session end.

★ THE CENTRAL TRADE-OFF — SODIUM LOADING

If the profile is not engineered for net-zero diffusive Na, the patient leaves the session salt-loaded. The downstream cascade is predictable and well-documented:

- Increased thirst between sessions.
- Higher interdialytic weight gain (IDWG).^{7 (#ref-7)}
- Worsened interdialytic hypertension.
- *Masked chronic volume overload* — pre-HD BP creeps up and the dry-weight drifts wrong-ward; LVH and HF accelerate.

This is the KDIGO Controversies Conference harm signal: dialysate Na individualization carries a meaningful prescribing-error risk and an indirect signal of harm when imprecisely applied. **Avoid sodium loading during dialysis** is the conference's explicit caution.^{1 (#ref-1)}

The safer design — sodium-balance-neutral profiling

The Song et al. (JASN 2005) data and the broader Na-balance literature show that profiles engineered for **net-zero diffusive Na transfer** can preserve the hemodynamic benefit while minimizing the IDWG penalty.^{8 (#ref-8)} The operative variable is not "start Na" — it is the **time-averaged dialysate Na** relative to the patient's pre-dialysis serum Na.

PREFERRED IMPLEMENTATION

Time-averaged dialysate Na $\approx$ pre-HD plasma Na. A profile that starts at 155 mEq/L and ends at 137 mEq/L over a 4-hr session has a time-averaged Na of ≈ 146 mEq/L — and is net-positive Na transfer in a patient with pre-HD Na of 138. Either lower the time-averaged Na (shift the whole profile down), or shorten the high-Na phase, until the gradient closes.

Prescription notes

- Specify **start Na, end Na, profile shape, session duration, and time-averaged dialysate Na** in the order — not just the bookend values.
- Anchor to the patient's **pre-HD plasma Na**. Without it, you are profile-blind.
- **Do not initiate** in patients with high IDWG, poorly controlled interdialytic hypertension, or known dietary salt non-adherence — these are precisely the patients in whom the technique will magnify harm.

- Pair every Na-ramping prescription with structured patient sodium-restriction counselling. Ramping is *not* a license to liberalize salt.
- Set an explicit reassessment interval (e.g. 4–6 sessions) and stopping rules tied to IDWG and pre-HD BP (§8).

EVIDENCE AT A GLANCE — SODIUM RAMPING

STRENGTH OF EVIDENCE

Multiple older RCTs and crossover studies; one 2024 RCT (combined Na + A/D-UF); KDIGO consensus framing.

WHAT IS NOT PROVEN

Mortality or hospitalization benefit.

WHAT IS PROVEN

Reduced IDH episodes and intradialytic symptoms; better post-dialysis BP maintenance in IDH-prone patients. [6](#) [7](#) [10](#) [11](#) [11](#)

PRINCIPAL HARM

Sodium loading → IDWG, interdialytic HTN, masked chronic volume overload. [1](#) [7](#)

§4 · TECHNIQUE — UF RAMPING

Ultrafiltration Profiling & UF Ramping

Time-varying UFR within a session. Reshapes the UFR curve to match the falling refill curve — but does not lower the average dose.

Figure 4. Profile-shape catalogue for sodium (top) and UFR (bottom). Same dose, different distribution. The operative variables are TIME-AVERAGED dialysate Na and MEAN UFR — both must be calculated, both must be documented (see §3 and §5).

Definition

UF profiling delivers a time-varying UFR within a single session, in contrast to constant UFR. Common shapes:

- **Descending** — UFR is highest in the first third (when refill reserve is highest) and tapers toward the end. The most physiologically defensible default.
- **Step** — discrete tiers, often three.
- **Ascending-descending (A/D)** — paired with sodium profiling in the 2024 RCT data.¹¹ (#ref-11)

Mechanism

The instantaneous plasma-refilling rate is highest at the start of the session and falls progressively as the interstitium empties. A descending UF profile *matches* the UFR to that curve: remove the most water when refill capacity is greatest, and protect the late-session window where refilling has collapsed and IDH events cluster.² (#ref-2)

Honest evidence framing

The Flythe crossover RCT is the most important piece of evidence to represent accurately. UF profiling **did** reduce intradialytic lightheadedness and reduced post-dialysis hypervolemia — but it **did not** reduce treatment-induced cardiac stress.¹³ (#ref-13) The greatest documented hemodynamic benefit appears when UF profiling is *combined* with sodium profiling.⁹ (#ref-9),¹¹ (#ref-11)

DISTRIBUTION ≠ DOSE

UF profiling reshapes the UFR curve. It does *not* reduce the **average** UFR. A patient whose mean UFR exceeds their safe ceiling cannot be rescued by profiling — they need **longer time, increased frequency, or a dry-weight reassessment**. Profiling that masks an unsafe

average is harm, not care.

Relationship to weight-based UFR limits

Profiling is about *distribution*; the *average* still matters. Anchor to the literature: the ≤ 13 mL/kg/hr threshold (and the more conservative 10 mL/kg/hr inflection signal) marks the population-level IDH risk transition, reaffirmed by 2025 reviews placing UF and IDWG as the dominant modifiable drivers of IDH. ^{14 (#ref-14),15 (#ref-15),18 (#ref-18)}

However — and this is the critical caveat the guide must carry — **chasing UFR limits *in isolation* can cause under-removal and volume expansion**. The 2018 review (PMID 29885084) and KDIGO both warn against treating any single UFR cutoff as an absolute rule when the alternative is leaving the patient wet. ^{16 (#ref-16)} The right framing: the UFR limit is an alarm to *reconsider time/frequency*, not a license to abandon volume control.

EVIDENCE AT A GLANCE — UF PROFILING

STRENGTH OF EVIDENCE

Crossover RCTs (mostly small); strongest signal in combined Na + UF designs.

WHAT IS NOT PROVEN

Reduced cardiac stress in isolation; mortality benefit. ^{13 (#ref-13)}

WHAT IS PROVEN

Reduced intradialytic lightheadedness; lower post-dialysis hypervolemia. ^{9 (#ref-9),13 (#ref-13)}

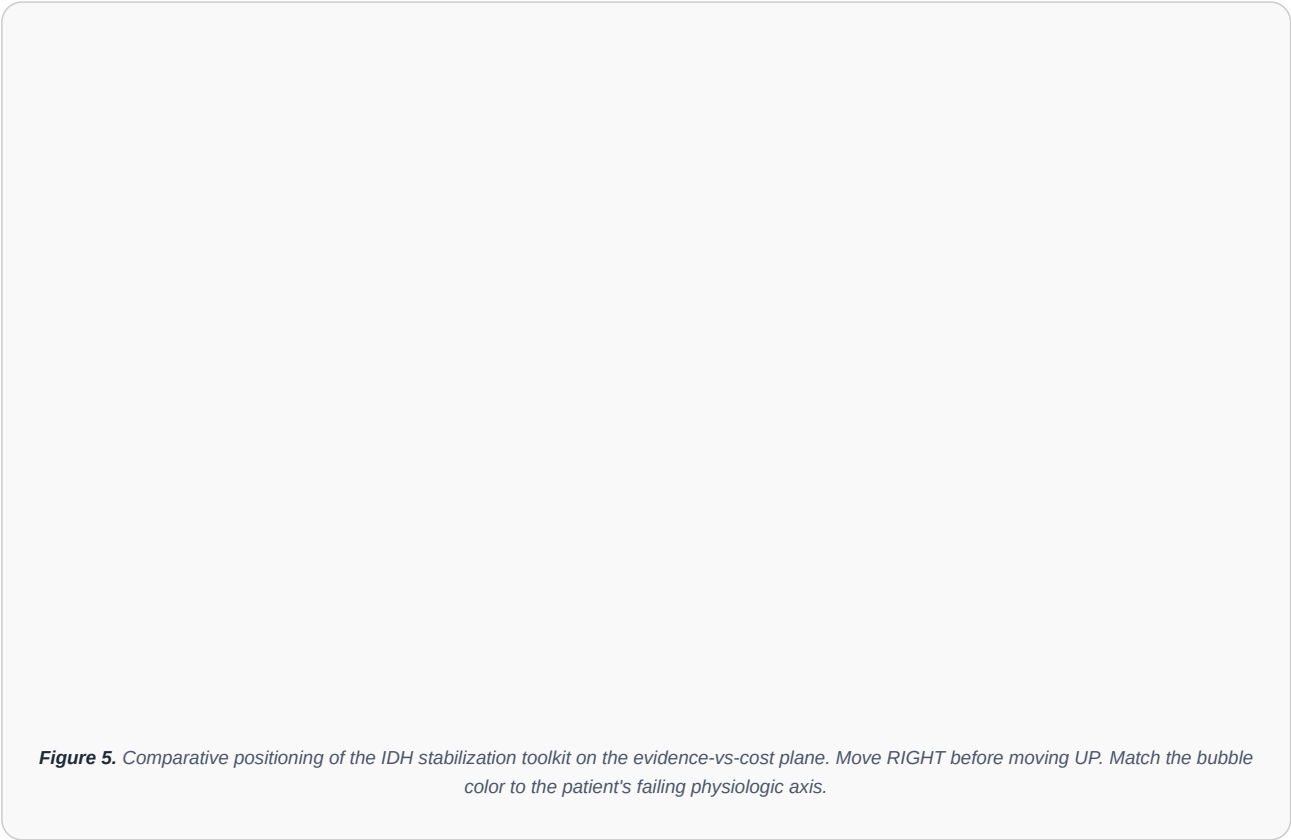
PRINCIPAL HARM

Masking an unsafe *mean* UFR; false reassurance that volume is controlled. ^{16 (#ref-16)}

S5 · COMPARISON

How These Compare to (and Combine with) the Alternatives

Cool dialysate and dry-weight/time optimization are higher-evidence first moves. The ramping/iso-UF toolkit is layered onto the genuinely refractory or fragile subset.



Intervention	Mechanism	Primary indication	Evidence	Principal harm / limitation	Adequacy impact	Resource / equipment
ISO-UF Sequential UF → HD	Preserves plasma osmolality & PVR during the UF phase.	Osmolar-collapse-driven IDH, large per-session solute shifts.	Small mechanistic RCTs + KDIGO framing.	Zero clearance in UF phase → adequacy trade-off; longer session.	Negative if HD block not compensated.	Any modern HD machine.
NA RAMPING (balance-neutral)	Replaces lost tonicity early; supports refilling & vascular tone.	IDH-prone with tolerable IDWG, refractory to first-line measures.	Multiple small RCTs / crossover; 2024 combined RCT.	Sodium loading → IDWG, interdialytic HTN, masked volume overload (KDIGO).	Neutral if net-zero Na.	Programmable Na on machine.
UF PROFILING Descending / A/D	Matches UFR to falling refill	Late-session IDH clustering with	Crossover RCTs	Masks unsafe	Neutral.	Programmable UFR profile.

Intervention	Mechanism	Primary indication	Evidence	Principal harm / limitation	Adequacy impact	Resource / equipment
	curve.	acceptable mean UFR.	(symptom reduction; cardiac stress unchanged).	<i>mean</i> UFR if used as cover.		
COOL DIALYSATE 35.5°C	Sympathetic activation; thermal vasoconstriction.	IDH-prone; co-first-line non-pharmacologic.	Cochrane (2019) — consistent IDH reduction. ¹⁹ (#ref-19)	Patient thermal discomfort.	Neutral.	Any HD machine.
BVM / BIOFEEDBACK UFC	Real-time refill-guided UFR adjustment.	Frequent, unpredictable IDH where staffing reactive control is limited.	Crossover comparative (2025). ¹⁷ (#ref-17)	Cost; equipment availability; mixed clinical signal.	Neutral.	BVM-capable machine + sensor.
Treatment time / frequency increase	Lowers required mean UFR per session.	Anyone with mean UFR > safe ceiling.	Strong observational; some RCT (FHN).	Patient acceptance; chair-time logistics; cost.	Improves.	Operational (chair time).
Dry-weight reassessment	Corrects systematic over-prescription of UF.	Suspected wrong-set dry weight (most IDH starts here).	Strong — BIA-supported. ²⁰ (#ref-20)	Requires repeated bedside assessment; BIA cost.	Neutral.	Clinical / BIA / lung US.
Antihypertensive timing review	Reduces pre-HD vasodilatation.	Any IDH-prone patient on multiple antihypertensives.	Pragmatic.	Misses interdialytic BP control if too liberal.	Neutral.	Operational.
Midodrine	α-1 vasoconstriction.	Refractory IDH despite the above.	Small studies; KDIGO Conference: insufficient. ¹ (#ref-1)	Supine HTN; cost; QID dosing.	Neutral.	Pharmacy.

SEQUENCE OF MOVES — PHILIPPINES DIALYSIS-UNIT REALITY

In a resource-aware unit, the rational order is: **1.** dry-weight reassessment + treatment-time review · **2.** cool dialysate (high-evidence, zero cost) · **3.** antihypertensive timing and IDWG counselling · **4.** UF profiling for documented late-session IDH · **5.** sodium-balance-neutral ramping if IDWG is tolerable · **6.** iso-UF / sequential dialysis for the osmolar-collapse phenotype · **7.** BVM-UFC if available and the above fails. Pharmacologic measures (midodrine) are layered on the residual refractory tail, not used early as a shortcut.

★ §6 · PATIENT SELECTION — THE CORE DELIVERABLE

Patient Selection Framework

The clinician must be able to identify the subset of patients these methods will have the greatest impact on — and the subset in whom they will cause harm.

Figure 2. Patient-selection algorithm. Five gating prerequisites → four-branch tool-to-phenotype matrix → three red de-selection nodes. Use as a session-prep checklist, not a population default.

7.1 — The high-yield phenotype (who benefits most)

INDICATIONS TO ENRICH CONSIDERATION

- **Recurrent symptomatic IDH** despite an achieved or realistic dry weight and acceptable IDWG ($\leq 3\text{--}4\%$ of body weight, ideally <1 kg/day on the long interdialytic gap).
- **Hemodynamic fragility** — autonomic dysfunction (long-standing diabetes), elderly, low cardiac reserve (diastolic dysfunction, LVH, EF $<40\%$), severe aortic stenosis, antihypertensive-dependent.
- **Low refill reserve / high ECW signatures on BIA** — high ECW/TBW ratio, low phase angle, high overhydration index. These are validated predictors of IDH risk.^{20 (#ref-20)}
- **Necessarily high per-session UF burden** — patients with IDWG that cannot be reduced in the short term, or a long interdialytic gap (post-Sunday HD in PH practice), where UF *distribution* can be optimized even when the average cannot immediately drop.
- **"Difficult-to-dry-weight" patients** — those who hit IDH before reaching euvolemia. Iso-UF / profiling buys the hemodynamic room to remove more fluid per session.

7.2 — Tool-to-phenotype matching matrix

Clinical driver	Preferred technique	Rationale	Caveat
Osmolar-collapse–driven IDH; large solute shifts (e.g. urea reduction $>75\%$, hyponatremic pre-HD Na).	ISO-UF (SEQUENTIAL)	Preserves tonicity & PVR during the UF block.	Adequacy trade-off; longer total time.

Clinical driver	Preferred technique	Rationale	Caveat
Late-session IDH clustering with adequate early tolerance; mean UFR within safe ceiling.	DESCENDING UF PROFILE	Matches UFR to falling refill curve.	Average UFR must still be safe.
IDH-prone with <i>tolerable</i> IDWG, refractory to cool dialysate & UF profiling alone.	NA-BALANCE-NEUTRAL RAMPING ± UF PROFILE	Tonicity replacement + refill optimization.	Strict net-zero Na; not for high-IDWG patients.
Frequent unpredictable IDH; BVM-capable machine available.	BVM-UFC (BIOFEEDBACK)	Real-time refill-guided UF rate adjustment.	Cost; evidence mixed; not substitute for selection.
Any IDH-prone patient, low resource setting.	COOL DIALYSATE FIRST	High-evidence, near-zero cost.	Thermal discomfort in some.

7.3 — Who to AVOID these in

DE-SELECTION LIST — DO NOT PRESCRIBE

- **High IDWG, interdialytic hypertension, or known dietary salt non-adherence.** Na ramping in this patient will worsen chronic volume overload — the patient who most needs *less* salt gets *more*.
- **Patients whose IDH is actually under-dialyzed or over-dry-weight error.** The IDH is the symptom; the wrong dry weight (or insufficient time) is the cause. Fix it first. Do not paper over volume overload with sodium loading.
- **Routine, hemodynamically stable patients.** No benefit demonstrated; adds cost, complexity, and prescribing-error risk (KDIGO).¹ (#ref-1)
- **Mortality-benefit expectation.** If the request is "let's add this to extend life expectancy," the answer is: *no* — *unproven*. Match the technique to the symptom it actually addresses.

7.4 — Pre-implementation checklist (clinician sign-off)

SIGN OFF ALL SIX BEFORE PRESCRIBING

- 1 **Dry weight reassessed** — clinical signs (pre-HD JVP, lung auscultation, pedal edema) ± BIA (overhydration index, ECW/TBW) ± lung ultrasound B-line count.
- 2 **Treatment time / frequency optimized** — would adding 30 minutes or moving to 4×/week solve this without a new prescription?
- 3 **IDWG addressed** — pre-HD weight reviewed for the last 6 sessions; salt-restriction counselling delivered (and documented).
- 4 **Cool dialysate trialed** — at 35.5°C for ≥4 sessions before adding any of the techniques in this guide.
- 5 **Antihypertensive timing reviewed** — long-acting agents moved off the morning of HD; mid-day doses considered.
- 6 **If proceeding** — exact prescription parameters defined, monitoring plan written (§8), reassessment date set (4–6 sessions out).

S7 · PRESCRIPTION WORKSHEETS

Prescription Worksheets (copy-ready)

Order-sheet templates with embedded safety guardrails. Transcribe into the HD prescription; do not initial off-template.

● RX 1 · ISO-UF / SEQUENTIAL DIALYSIS

Modality	Sequential dialysis: iso-UF block → conventional HD block
Iso-UF duration	_____ min (individualized; no guideline standard – document rationale)
Target UF in iso-UF phase	_____ L (typically 40–60% of total session UF, front-loaded)
Iso-UF phase blood flow	_____ mL/min
Dialysate flow (iso-UF)	OFF
HD phase duration	_____ min
HD phase Qb / Qd / Kt/V target	_____ / _____ / ≥ 1.4 (single pool)
Dialysate composition	Na 138 · K _____ · Ca 1.25 · HCO ₃ 32–35
Dialysate temperature	35.5 °C (cool)
Anticoagulation	Heparin to cover BOTH phases (verify circuit clotting check)
Total UF / total time	_____ L over _____ min
Reassess in	6 sessions – IDH count, post-HD orthostatics, achieved Kt/V

Guardrails. Mean UFR over the WHOLE session $\leq$ patient-specific ceiling (≤ 13 mL/kg/hr; consider 10 mL/kg/hr in cardiac fragility). If HD-block Kt/V < 1.2 , extend HD block or move to longer session — do not accept under-dialysis to keep the iso-UF block.

● RX 2 · SODIUM RAMPING (BALANCE-NEUTRAL)

Pre-HD plasma Na (anchor)	_____ mEq/L (REQUIRED before prescribing)
Profile shape	☐ Step ☐ Linear descending ☐ A/D (paired with UF profile)
Start dialysate Na	_____ mEq/L (typical 148–152; avoid >155 unless osmolar-collapse phenotype documented)
End dialysate Na	_____ mEq/L (typical 136–138)
Session duration	_____ min
Time-averaged dialysate Na	_____ mEq/L (must $\approx$ pre-HD plasma Na ± 1 – the OPERATIVE variable)
Net diffusive Na target	Net-zero (± 0.3 mmol/L per kg body water)
IDWG ceiling (eligibility)	$\leq 3\%$ body weight or ≤ 2.5 kg, whichever lower
Patient counselling	Salt restriction reinforced and documented THIS visit
Reassess in	4 sessions – IDWG trend, pre-HD BP trend, IDH count

Guardrails. If IDWG rises >0.5 kg above baseline OR pre-HD systolic BP rises >10 mmHg over 4 sessions → STOP profile (sodium loading detected). Recheck plasma Na before next prescription cycle.

● RX 3 · UF PROFILING / RAMPING

Post-HD dry weight	_____ kg
Total session UF	_____ L
Session duration	_____ min
Mean UFR (calculated)	_____ mL/kg/hr (target ≤ 13 ; aim ≤ 10 in cardiac fragility)
Profile shape	☐ Descending (linear) ☐ Step (3 tiers) ☐ A/D paired with Na ramp
Peak UFR (first third)	_____ mL/hr ($\leq 1.4 \times$ mean UFR)
Late-session UFR floor	_____ mL/hr ($\geq 0.5 \times$ mean UFR; do not drop to zero)
Dialysate temperature	35.5 °C (pair with cool dialysate)
Reassess in	6 sessions – late-session SBP nadir, post-HD orthostatics, achieved dry weight

Guardrails. Profiling does NOT lower the mean UFR. If mean UFR exceeds the patient's safe ceiling: extend session time or increase frequency BEFORE adding a profile. Do not use profiling to "rescue" a structurally unsafe prescription.

SINGLE-SOURCE-OF-TRUTH PRINCIPLE

The order sheet should record the patient's **pre-HD plasma Na**, the **mean UFR (mL/kg/hr) check**, and the **reassessment date** as required fields. These three values prevent the three most common errors: profile-blind Na prescribing, distribution-without-dose UF profiling, and indefinitely continued ramping after the precipitant has resolved.

S8 · MONITORING & EXIT

Monitoring, De-Prescribing & Reassessment

These are temporizing tools. Every prescription needs a written exit.

What to track

Metric	Why	Cadence	Action threshold
Per-session SBP nadir & IDH events	Primary efficacy endpoint of any prescription in this guide.	Every session.	No reduction over 6 sessions → de-escalate or change tool.
Pre-HD BP trend	Detects creeping volume overload from Na loading.	Every session; review trend at 4-session intervals.	Rise >10 mmHg SBP over 4 sessions → stop Na ramping; reassess dry weight.
IDWG trend	Earliest signal of sodium loading.	Every session.	Rise >0.5 kg above baseline → stop Na ramping.
Post-HD orthostatics	Captures residual hemodynamic vulnerability after the technique is added.	Every 4 sessions.	Persistent orthostatic SBP fall >20 mmHg → mean UFR review; reduce dose.
Achieved vs target dry weight	Confirms profiling is enabling — not masking — adequate volume control.	Monthly clinical + 3-monthly BIA if available.	Drift upward → reassess dry weight, not just the prescription.
Residual kidney function (urine output)	RKF is a prognostic asset; IDH and excessive UF erode it.	Monthly.	Falling RKF → escalate stability strategy and reduce UF stress.
Achieved Kt/V (single-pool)	Iso-UF risk: under-dialysis if HD block does not compensate.	Monthly.	Kt/V < 1.2 → extend HD block or move to longer session.

Stopping rules — build the exit in

STOP THE PRESCRIPTION IF ANY OF THESE APPLY

- **Pre-HD BP rising** >10 mmHg systolic over 4 sessions on Na ramping (sodium loading detected).
- **IDWG rising** >0.5 kg above baseline on Na ramping (same).
- **No reduction in IDH count** after 6 sessions of a given prescription — the right answer is not a higher dose, it is a different tool or a different problem.
- **The precipitant has resolved** — intercurrent infection treated, mis-set dry weight corrected, antihypertensive timing fixed. The technique should de-escalate too.
- **Iso-UF block undermining Kt/V** — extend session or de-escalate to UF profiling.

KDIGO BP / volume anchors

Cross-link to the KDIGO 2020 Controversies Conference framework already used in the unit: avoid sodium loading, individualize on the basis of pre-HD plasma Na, and pair every intradialytic stabilization technique with structured volume reassessment. ^{1 (#ref-1)}

S9 · PITFALLS & MYTHS

Pitfalls & Myths — clinician quick-hits

The five sentences that, said out loud at handover, prevent the five most common errors.

MYTH

“Sodium ramping is a free win — it makes IDH go away.”

Truth: Non-neutral ramping buys intradialytic stability at the cost of chronic volume burden — higher IDWG, interdialytic hypertension, masked dry-weight drift. The KDIGO Conference explicitly cautions against sodium loading during dialysis.¹ (#ref-1)

MYTH

“UF profiling fixes a high average UFR.”

Truth: Profiling reshapes the curve; it does not lower the area under it. A patient whose *mean* UFR exceeds their safe ceiling needs longer time, more frequency, or a corrected dry weight — not a profile.¹³ (#ref-13)

MYTH

“Iso-UF improves dialysis clearance.”

Truth: Iso-UF delivers zero clearance during the UF phase. Total session Kt/V is the HD-phase Kt/V alone. The adequacy trade-off is real and must be planned for.

MYTH

“These techniques reduce mortality.”

Truth: Unproven. The honest claim is symptom and IDH reduction in selected patients. Use them for that, not as life-prolonging interventions.¹ (#ref-1)

MYTH

“The 13 mL/kg/hr UFR limit is an absolute rule.”

Truth: Chasing UFR cutoffs in isolation can drive under-removal and volume expansion. The threshold is a *signal to reconsider time/frequency*, not a license to send the patient home wet.¹⁶ (#ref-16)

§10 · REFERENCES

References

Cited inline; full Vancouver list also rendered in the closing accordion at the end of the page.

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